

HA VAN DUY

Senior Data Engineer | Data Platform

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PROFESSIONAL SUMMARY

Senior Data Engineer with 4+ years building production-grade data platforms at MoMo (Vietnam's leading fintech super-app, 30M+ users). Architect of the company's lakehouse migration, its SOC 2-compliant access management and data-protection stack, and an Agentic-AI platform now serving 30+ business workflows and 100K+ executions. Owns systems end to end — from ingestion and serving to governance, security, and AI-assisted automation for business users. Google Cloud Certified Professional Data Engineer, co-author of two peer-reviewed ICCCI papers, and open-source contributor (SQLGlott, Great Expectations).

CORE SKILLS

Languages: Python, SQL, Java, C++, R, Bash

Big Data & Lakehouse: Apache Spark, Trino/Presto, BigQuery, Delta Lake, Apache Iceberg, dbt

Orchestration & Infra: Apache Airflow, Kubernetes (GKE & on-prem), Docker, Bazel, Helm

Cloud: Google Cloud (BigQuery, GCS, Dataproc, GKE, Cloud Functions, Identity), Oracle (APEX, OCI)

AI / ML Engineering: Generative AI, Agentic AI design, LangChain, Claude (Cowork, Code), RAG, prompt engineering, evaluation pipelines

Data Governance & Security: Apache Gravitino, credential vending, Apache Ranger, DataHub, Great Expectations, SOC 2 controls, OAuth2 / OpenID Connect, PII auto-classification

MLOps & DevOps: CI/CD (GitHub Actions, GitLab CI, Jenkins), monitoring, IaC patterns, on-call data platform ops

Languages (Human): Vietnamese (native), English (Intermediate – reading/writing technical docs daily)

CERTIFICATIONS

Professional Data Engineer — Google Cloud

Issued Aug 14, 2026 · Expires Aug 14, 2028

PROFESSIONAL EXPERIENCE

MoMo — Data Platform, Technology Center

Ho Chi Minh City

Senior Data Engineer (Apr 2026) | Data Engineer II | Junior Data Engineer | Trainee, MoMo Talents Jan 2022 – Present

- **Lakehouse architect & migration owner:** designed and rolled out a Trino + Iceberg/Delta on GCS lakehouse running on GKE; migrated core ETL workloads off BigQuery and cut up to **70% cost per workload** using spot instances with no SLA breaches.
- **SQL transpiler & migration tooling:** built an end-to-end transpiler on top of SQLGlott to translate BigQuery → Trino and Spark dialects, reducing human migration effort by **~90%** across hundreds of pipelines.
- **Cost optimization at platform scale:** led query optimization and BigQuery slot allocation; delivered **40% GCP cost reduction** across all departments with zero workload disruption.
- **Access Management platform (SOC 2-compliant):** shipped a centralized, time-based privileged access tool handling **100% of data-access requests** across warehouses, lakehouses, and services, plus a Google Chat bot for instant approval and execution.
- **Data protection – unified catalog & credential vending (2026):** rolled out Apache Gravitino as the single metadata catalog for the whole lakehouse, then shipped credential vending to issue short-lived, scoped storage tokens at runtime — retiring every long-lived, over-privileged service account from the Spark compute engines.
- **Data security – PII auto-classification:** deployed auto-scanning and tagging over the lakehouse to detect and protect sensitive personal data, enforce least-privilege access, and meet regulatory compliance.

- **Data Agent / Agentic AI platform:** built a scalable Agentic-AI framework that now powers **30+ AI workflows** with **100K+ total runs**, processing **10B+ tokens** and **3K+ weekly executions**; ships AI-generated daily and personal customer-feedback alerts to every business unit via Google Chat.

SELECTED PROJECTS

Agentic AI Platform & Data Agent — MoMo

2025 – Present

- Built a scalable, maintainable framework that lets MoMo engineers ship new AI agents in days, not months; explored autonomous decision-making patterns in agentic design.
- Powers chatbots and AI-summarization for engineers and business users to query internal data/documents; clusters user feedback into key topics with metrics for fast business action.
- **Impact:** 80% reduction in engineering time on periodic data analysis; AI-generated daily insight reports delivered to every BU on Google Chat.
- **Tech:** Python, GenAI, Agentic AI patterns, LangChain, FastAPI, SMTP, Chatbots, Claude (Cowork, Code).

Access Management & Data Protection — MoMo

2024 – Present

- SOC 2-compliant platform with time-based privileged access, policy tagging, audit logs, and compliance reports across data warehouses, lakehouses, and internal services.
- Google Chat approval bot for instant request handling; integration with the new lakehouse and with other MoMo data platforms.
- Unified metadata catalog (Apache Gravitino) with credential vending: runtime-generated short-term storage tokens replace static service accounts on compute engines.
- **Tech:** SOC 2 controls, Apache Gravitino, credential vending, Apache Ranger, FastAPI, OAuth2 / OpenID Connect, Python.

Data Lakehouse & Pipeline Migration — MoMo

2023 – Present

- Researched and selected open table formats (Iceberg / Delta); designed Trino-on-GKE for distributed SQL on GCS; integrated lakehouse with new access management & governance stack.
- Migrated core ETL pipelines using an in-house SQLGlot-based transpiler. **Impact:** up to 70% lower cost per workload, 90% lower migration effort.
- **Tech:** Trino, Spark, Apache Ranger, GKE, GCS, BigQuery Storage, dbt, Airflow, SQLGlot.

Cost Optimization on GCP — MoMo

2022 – 2023

- Optimized queries, moved services/ETL/ELT pipelines to on-prem Kubernetes, ran a BigQuery → Vertica experiment, and centralized BigQuery resource allocation by team.
- **Impact:** 40% GCP cost saving across the platform with zero workload regressions.

PUBLICATIONS

MEP: A Comprehensive Medicines Extraction System on Prescriptions — ICCCI 2023, Computational Collective Intelligence, pp. 713–725.

Medical Prescription Recognition Using Heuristic Clustering and Similarity Search — ICCCI 2022, pp. 768–780.

OPEN-SOURCE CONTRIBUTIONS

- **SQLGlot** – Contributed improvements to BigQuery ↔ other-dialect translation accuracy; the same engine powers MoMo's internal transpiler.
- **Great Expectations** – Added Vertica dialect support so GX can validate data quality on Vertica databases.

EDUCATION

University of Science, VNU-HCM

Bachelor of Science, Data Science in Computer Science — GPA 8.5 / 10

Ho Chi Minh City

Sep 2018 – May 2022